

Continuous functions – 9

Approximation: Weierstrass, Bernstein, Stone

Homework

1 сентября 2026 года

Theoretical minimum. The Weierstrass approximation theorem; Bernstein polynomials and the estimate through the modulus of continuity; the Stone–Weierstrass theorem; density of piecewise-linear functions; the moment problem $\int f x^n = 0$; Chebyshev’s alternation theorem for the best uniform approximation; Müntz’s theorem.

1. Prove that $B_n f \rightarrow f$ uniformly for continuous f , that B_n preserves monotonicity and convexity, and that $B_n f = f$ exactly for affine f .
2. Prove that a continuous function on $[0, 1]$ with $\int_0^1 f(x)p(x)dx = 0$ for every polynomial p of degree at most n has at least $n + 1$ sign changes.
3. Prove Chebyshev’s alternation theorem: a polynomial of degree $\leq n$ is the best uniform approximation to f on $[a, b]$ if and only if the error attains its maximal modulus with alternating signs at $n + 2$ points.
4. Find the best uniform approximation of x^n by polynomials of degree $\leq n - 1$ on $[-1, 1]$ and explain the appearance of the Chebyshev polynomial.
5. Prove that the set of polynomials with rational coefficients is dense in $C[0, 1]$, and deduce that $C[0, 1]$ is separable.
6. Prove that the trigonometric polynomials are dense in the space of continuous 2π -periodic functions, using the Fejér kernel rather than Stone–Weierstrass.
7. Prove that a continuous function on $[0, 1]$ can be approximated uniformly by polynomials with integer coefficients if and only if $f(0)$ and $f(1)$ are integers.

8. Let f be continuous on $[0, \infty)$ with a finite limit at infinity. Prove that f can be uniformly approximated by finite linear combinations of the functions e^{-nx} , $n \geq 0$.
9. Prove that the algebra generated by $\{1, x^2, x^3\}$ is dense in $C[0, 1]$ but its linear span is not, and explain which hypothesis of Stone–Weierstrass is responsible.
10. Prove that for every continuous f on $[0, 1]$ and every $\varepsilon > 0$ there is a polynomial p with $\|f - p\|_\infty < \varepsilon$ and $p(0) = f(0)$, $p(1) = f(1)$, and show that one cannot in general also prescribe $p'(0)$.